
COURSE TOPICS

1. **Gateway Architecture:** Installation, module system, web management interface, unlimited licensing model, and backup/restore procedures.
2. **Tag System:** OPC, expression, and memory tag types; UDT templates for repeated equipment; tag database as the system foundation.
3. **OPC-UA Integration:** Ignition as OPC-UA client and server, device driver configuration, and PLC-to-live-tags workflow.
4. **Perspective Module:** Browser-based HMI architecture, component library, View/Page/Session structure, and tag binding without custom code.
5. **Scripting:** Jython on tag events, gateway events, and schedules; system library access to tags, databases, and alarm pipelines.
6. **Alarm Notification:** Alarm modes, notification pipelines, escalation rules, roster configuration, and email/SMS delivery infrastructure.
7. **Tag Historian:** Storage providers, partitioning strategy, aggregation modes, and Store and Forward for gap-free historical data.
8. **Database Integration:** Named queries, Transaction Groups, and the path from field sensor value to SQL table row.
9. **Security and Access Control:** Identity providers, role mapping, project permissions, and SSL/TLS configuration across single-site and multi-site deployments.
10. **Field Lab:** End-to-end SCADA build — OPC-UA device connection, tag configuration, Perspective dashboard, alarm pipeline, historian, management report.

LEARNING OUTCOMES

- Connect an OPC-UA device and expose live tags in a functioning Ignition project.
- Build a Perspective dashboard with live trend displays, alarm status, and navigation.
- Configure an alarm notification pipeline with escalation rules and SMS/email delivery.
- Write Jython scripts for tag event handling, scheduled tasks, and database queries.
- Query historian data and generate a management report from stored process values.

Certification Relevance

- **Inductive University Certified:** Core, Advanced, and Gold exams are free, self-paced, and industry-recognized; this course provides the structured foundation those exams assume.
- **ISA CCST:** SCADA system configuration, historian, and alarm management are tested topics at CCST Levels II and III.
- **Ignition Integrator:** Required for Inductive Automation Certified Integrators; demonstrates full-stack project delivery capability.

Next Steps

1. Download the Ignition trial from [inductiveautomation.com](https://www.inductiveautomation.com) (free, resets every two hours, fully featured and sufficient for the entire course).
2. Complete Inductive University Core certification (free, online, approximately 20 hours of video and exercises).
3. Build the Field Lab application from scratch using your own trial instance as the capstone.
4. Pursue Gold certification for the Certified Integrator pathway and higher-value project work.